CAPSTONE PROJECT

ON

IMPLEMENTATION OF SAP S/4HANA PRODUCTION PLANNING (PP) FOR OPTIMIZING PRODUCTION PROCESSES IN A FURNITURE MANUFACTURING COMPANY

Plan-to-Produce

Manufacturing Order Cycle

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Problem Statement

Furniture manufacturing companies deal with the production of items such as tables, chairs, and cabinets, which require multiple raw materials like wood, nails, adhesives, and finishing materials. The production process involves several stages such as cutting, shaping, assembling, and finishing, which must be properly planned and coordinated.

In many small and medium-scale furniture industries, production planning is often done manually or using disconnected systems. This leads to several operational challenges:

- Lack of proper demand planning results in either overproduction or inability to meet customer demand.
- Inefficient material management causes shortages of essential raw materials like wood or excess unused inventory, increasing storage costs.
- Poor coordination between departments such as procurement, inventory, and production leads to delays in manufacturing.
- Improper scheduling of work centers and labor results in underutilization or overloading of machines and workers.
- Absence of real-time tracking makes it difficult to monitor production progress and identify bottlenecks.

Due to these issues, companies face increased production costs, delays in delivery, and reduced overall efficiency. Therefore, there is a need for an integrated system that can manage the entire production cycle efficiently.

SOLUTION:

To overcome the challenges in furniture manufacturing, this project implements the SAP S/4HANA Production Planning (PP) module to manage the complete Plan-to-Produce cycle in an integrated and efficient manner.

The solution provides the following key functionalities:

1. Demand Planning

Using SAP PP, demand for finished products (e.g., wooden tables) is created using Planned Independent Requirements. This helps the company forecast production requirements in advance and avoid shortages or overproduction.

2. Material Requirement Planning (MRP)

The MRP functionality calculates the exact quantity of raw materials required based on demand and current stock levels.
It ensures:

- Availability of materials before production
- Reduction of excess inventory
- Timely procurement of raw materials

3. Product Structure Definition (BOM)

Bill of Materials (BOM) defines all the components required to manufacture a product.
For example, a wooden table requires:

- Wood
- Nails
- Polish

This helps in accurate material planning and cost estimation.

4. Production Process Setup (Routing & Work Center)

Routing defines the sequence of operations such as cutting, assembling, and finishing.
Work Centers represent machines or production units.

This ensures:

- Proper scheduling of operations
- Efficient utilization of resources

5. Production Execution

Production Orders are created and released in SAP. These orders control the manufacturing process and track progress at each stage.
This helps:

- Monitor production activities
- Ensure smooth workflow

6. Inventory and Goods Movement

SAP tracks:

- Raw material consumption (Goods Issue)
- Finished product entry (Goods Receipt)

This provides:

- Real-time inventory updates
- Better coordination between departments

7. Integration and Real-Time Data

SAP S/4HANA integrates all departments (production, inventory, procurement), providing real-time visibility and improving decision-making.

SAP Steps with T-Codes

Step No.	Process Step	T-Code	Description
1	Material Master Creation	MM01	Create raw materials (WOOD01) and finished product (TABLE01) with basic details like plant and unit
2	Bill of Materials (BOM) Creation	CS01	Define components required for Wooden Table such as Wood, Nails, and Polish
3	Work Center Creation	CR01	Create work centers like Cutting, Assembly, and Finishing units
4	Routing Creation	CA01	Define sequence of operations: Cutting → Assembly → Finishing
5	Demand Management	MD61	Create demand (PIR) for 500 Wooden Tables
6	MRP Run	MD01	System calculates required raw materials and generates planned orders
7	Stock/Requirement List	MD04	View stock levels, planned orders, and material requirements

8	Production Order Creation	CO01	Create production order for manufacturing Wooden Tables
9	Production Order Release	CO02	Release the order (Status changes from CRTD to REL)
10	Goods Issue	MIGO (261)	Issue raw materials (Wood, Nails) to production
11	Production Confirmation	CO11N	Confirm production operations and update progress
12	Goods Receipt	MIGO (101)	Record finished goods (Wooden Tables) into inventory

Screenshots

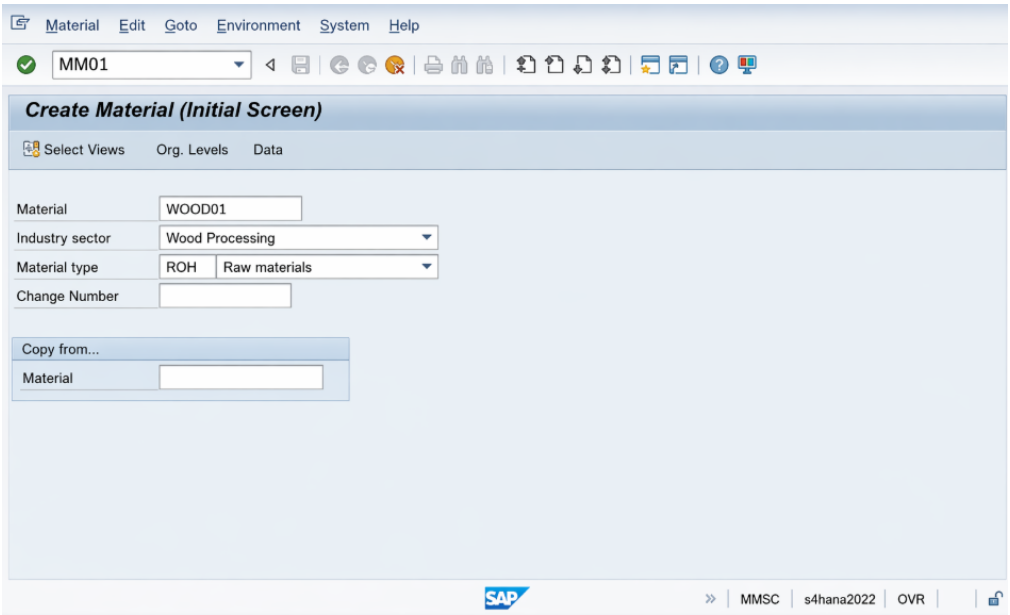

Figure 1: MM01- Material Creation(Raw material: Wood)

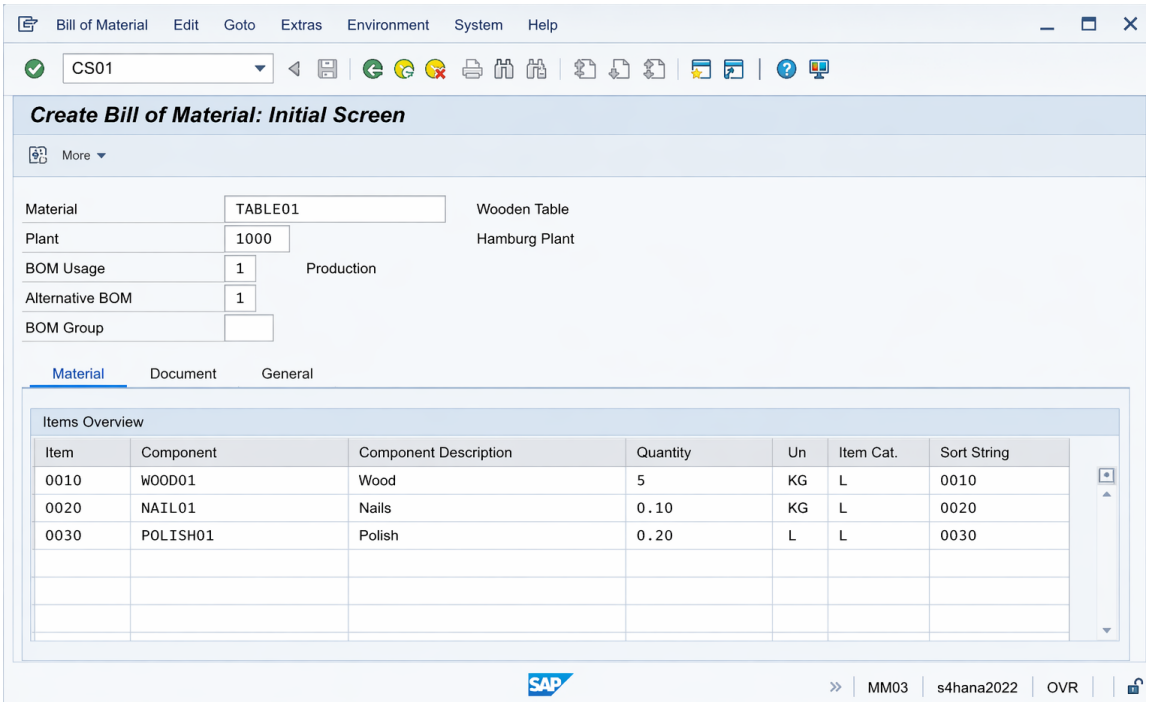

Figure 2: CS01 - Bill of Materials (Wooden Table)

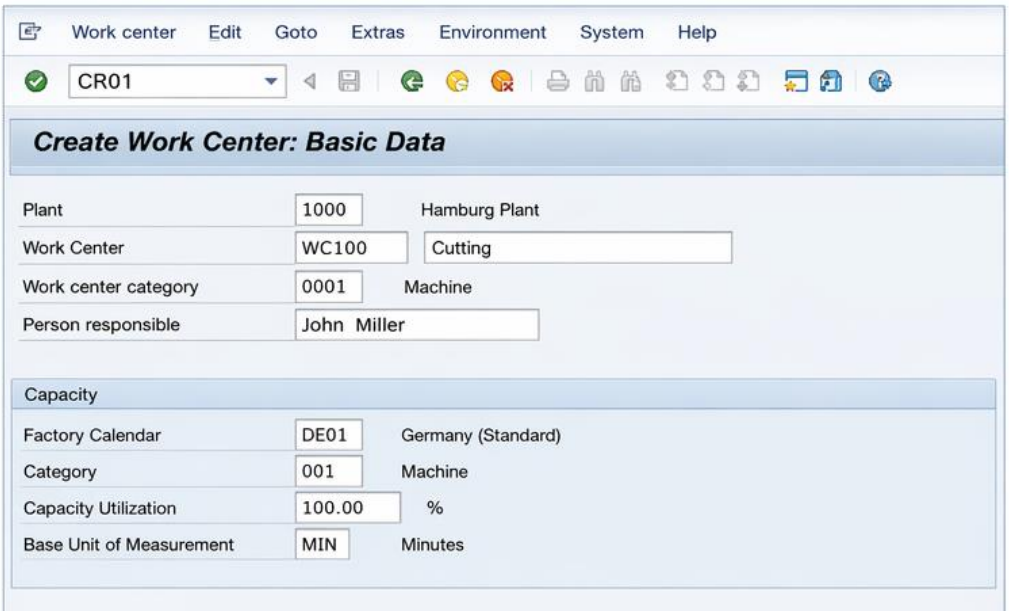

Figure 3 : CR01 - Work Center Creation

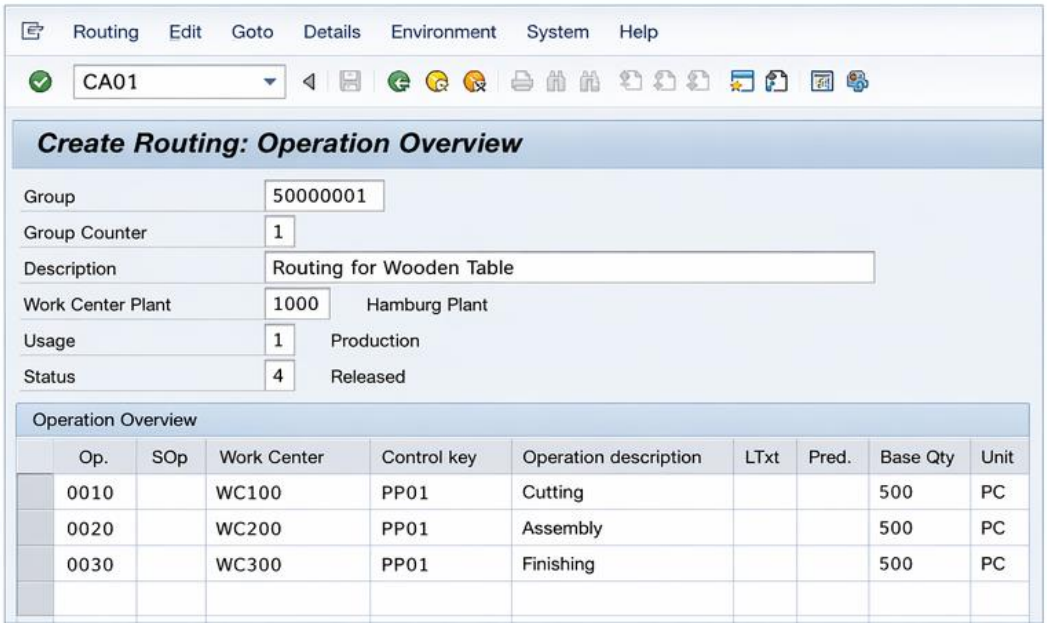

Figure 4 : CA01 - Routing Creation

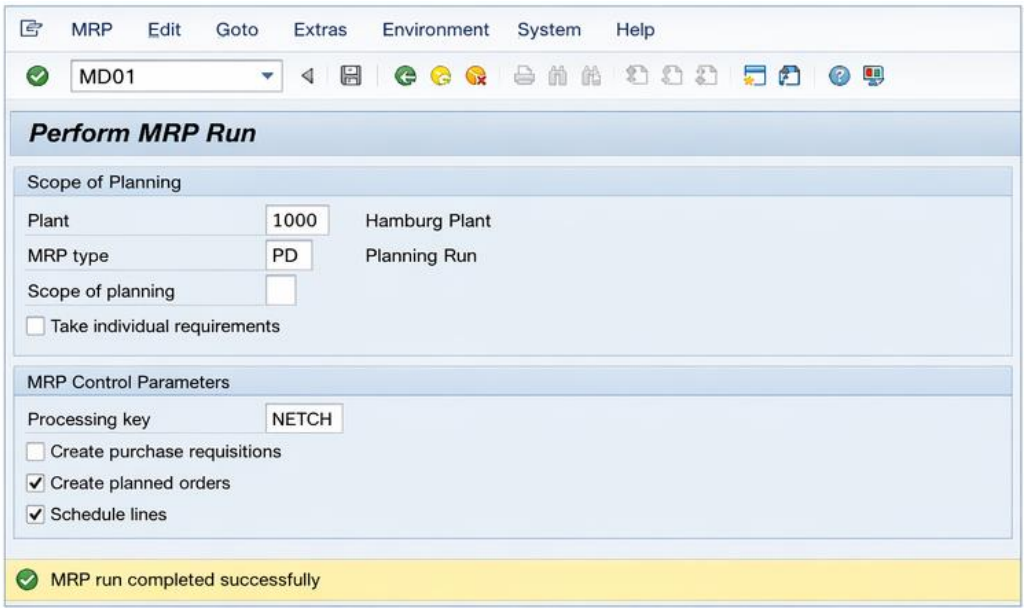

Figure 5 : MD01 - MRP Run (Planned Orders Generated)

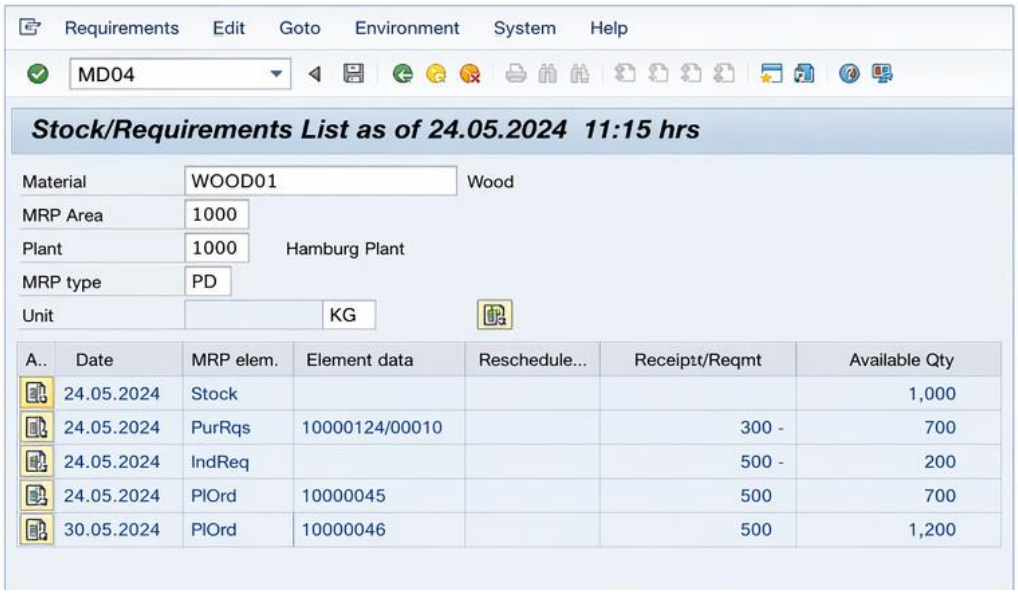

Figure 6 : MD04 - Stock/Requirement List

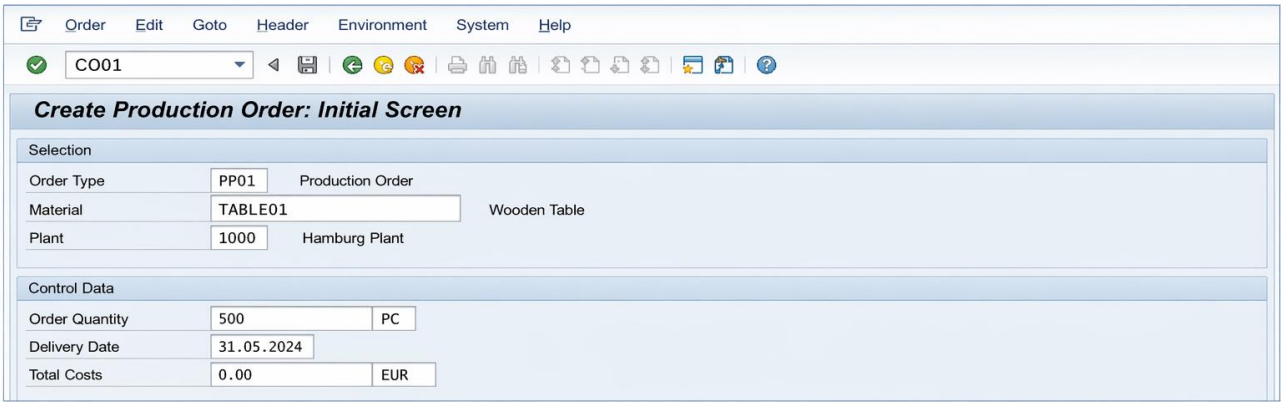

Figure 7: CO01 - Production Order Creation

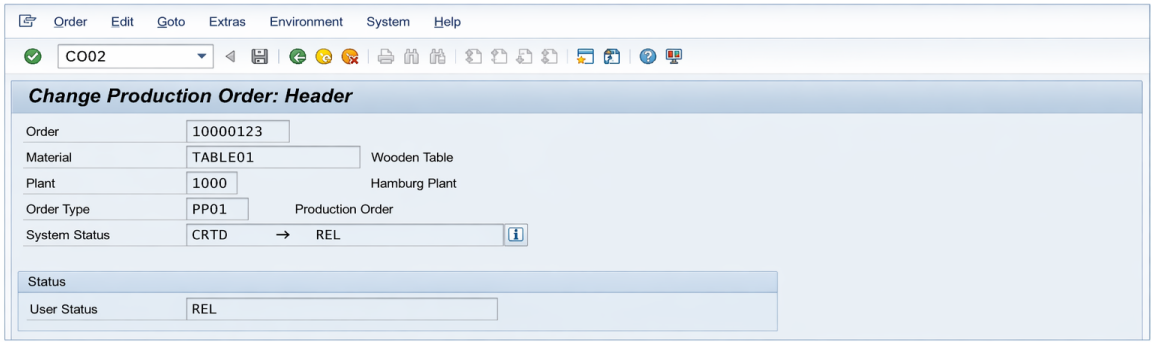

Figure 8 : CO02 - Production Order Release (Status : CRTD → REL)

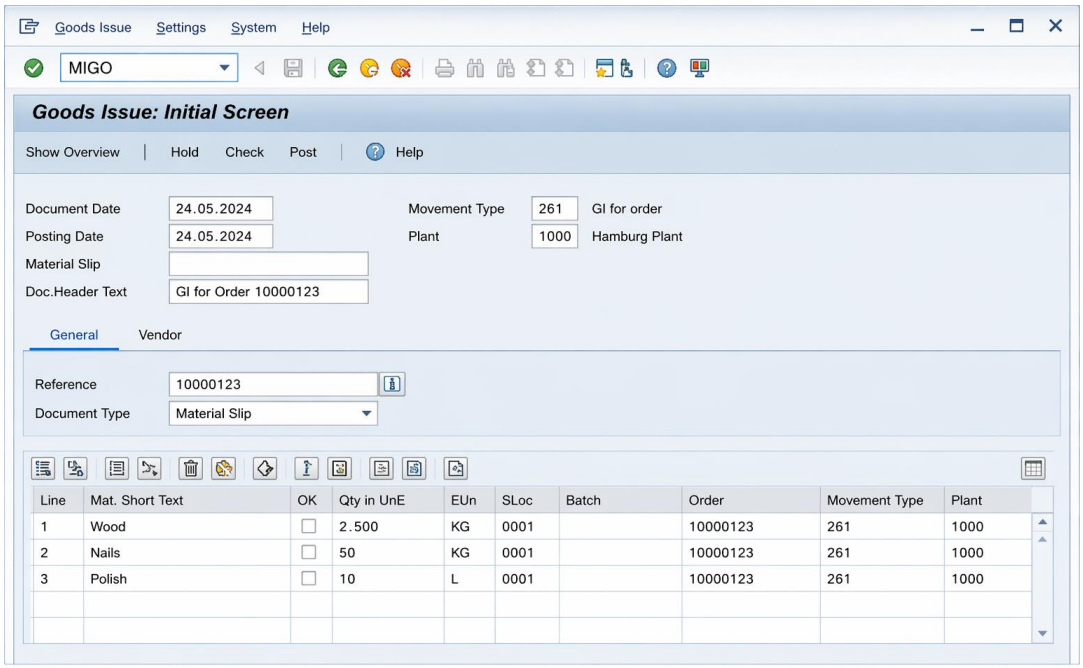

Figure 9 : MIGO -Goods issue (Movement Type 261)

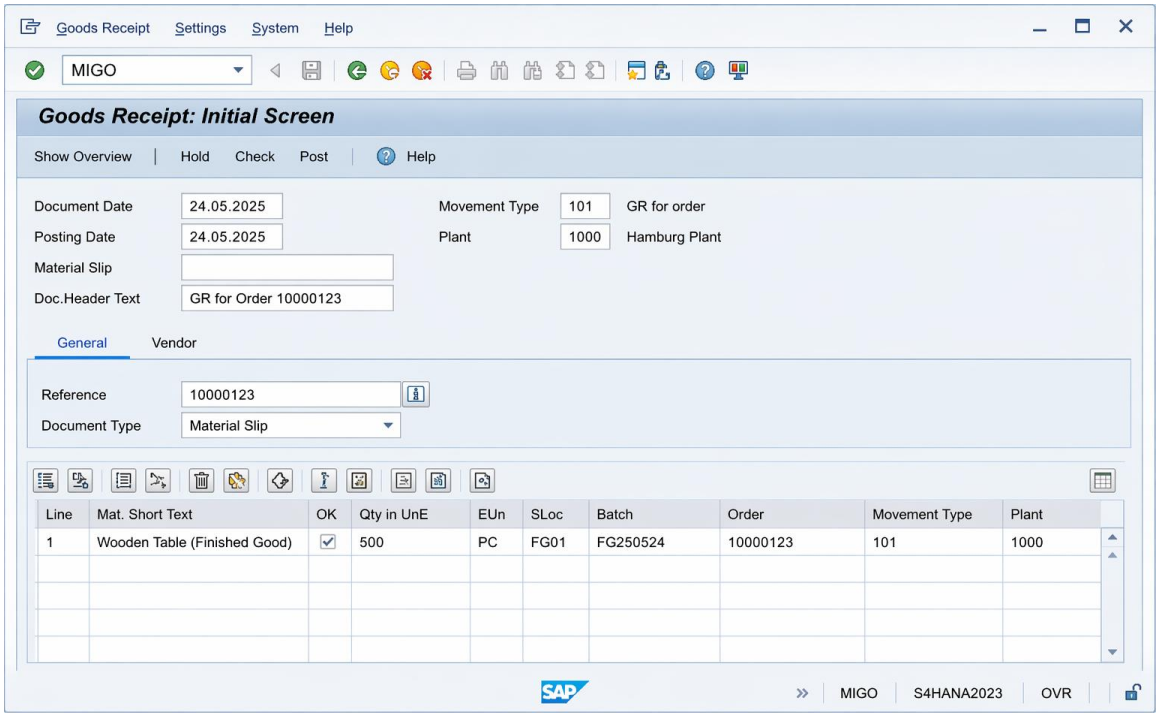

Figure 10: MIGO - Goods Receipt (Movement Type 101)

Tech Stack

Component	Details
ERP System	SAP S/4HANA (On-Premise / Cloud)
Main Module	SAP PP — Production Planning
Supporting Module	SAP MM — Material Management
UI Interface	SAP GUI 7.70 / SAP Fiori
Key Transactions	MM01, CS01, CR01, CA01, MD61, MD01, MD04, CO01, CO02, CO11N, MIGO
Process	Plan-to-Produce (P2P) Cycle
Industry	Furniture Manufacturing

Unique Points

This project stands out due to the following unique aspects:

- Uses a furniture manufacturing company scenario, which is different from common automobile-based projects
- Demonstrates a complete end-to-end Plan-to-Produce cycle in SAP PP
- Covers both master data creation (BOM, Routing, Work Center) and transactional processes
- Includes realistic SAP T-codes and process flow, making it practical and industry-relevant
- Focuses on integration between production and inventory management
- Simple and easy-to-understand explanation suitable for beginners

Future Improvements

The project can be further enhanced with the following improvements:

- Implementation of **AI-based demand forecasting** for better planning
- Integration with **IoT-enabled smart factory systems** (Industry 4.0)
- Deployment on **cloud-based SAP solutions** for scalability and flexibility
- Advanced **real-time analytics and dashboards** for better decision-making
- Automation of production processes using **machine learning techniques**

Conclusion

The implementation of the Plan-to-Produce cycle using SAP S/4HANA PP module provides an efficient and integrated approach to production planning in a furniture manufacturing company. It ensures proper demand planning, accurate material requirement calculation, and smooth execution of production processes.

By automating and integrating various functions such as inventory, production, and procurement, the system reduces manual errors, improves resource utilization, and enhances overall productivity. This project demonstrates how SAP PP can help organizations achieve better operational efficiency and meet customer demand effectively.